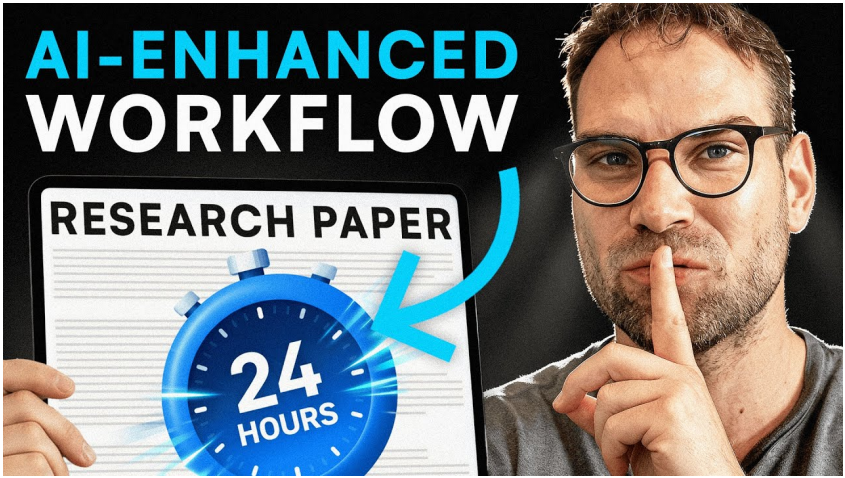
WHU
Otto Beisheim School of Management

Using AI for Reproducible Research

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A full research paper in 24 hours



Source: youtube.com/watch?v=YdpFu6Fkr58

What does research say (for the time being)?

AI handles the technical work better, but struggles with ideas and research judgment.

AI-written papers are weaker in ideas than in execution

AI models rated 912 AI-written economics papers and 41 published papers (Li, 2026)

They also rated scored equally high on execution, but fewer ideas reached top-journal level

Authors reject the work produced by AI agents working on the same RQs

AI agents had six days to answer the question of an unpublished paper (Kirgis et al., 2026)

The paper's authors reviewed the result: the conclusions did not follow from the evidence

What happens when experts work with an AI agent?

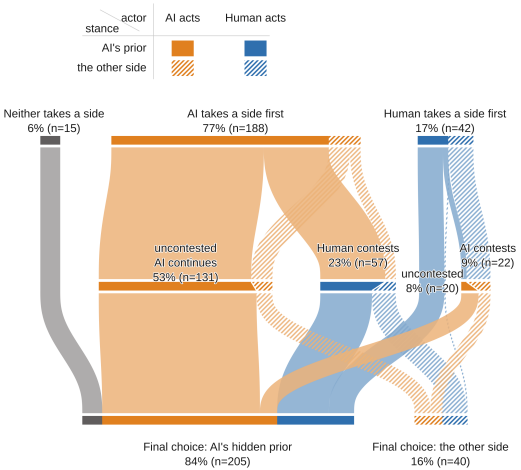


Figure 4 in Herdağdelen et al. (2026): all five decisions (245 in total), who took a side first, and whose side the final choice took

50 causal-inference experts analyzed data with Claude Code

A coin flip set which option the agent leaned toward on four analytical decisions

The experts went with the agent's lean 86% of the time (50% by chance)

Over half still rated the recommendation as more their own than the agent's

An Anthropic report, not peer reviewed: one task, one agent, and a lean that the system prompt amplified

Do workers hide AI use from employers?



Ivanti surveyed over 6,000 office workers and 1,200 IT professionals in February 2025 (Ivanti, 2025)

Of those who use generative AI at work, 32% keep their use of some AI tools a secret from their employer

They name the “secret advantage” (36%) and the worry that their job may be cut (30%) as reasons

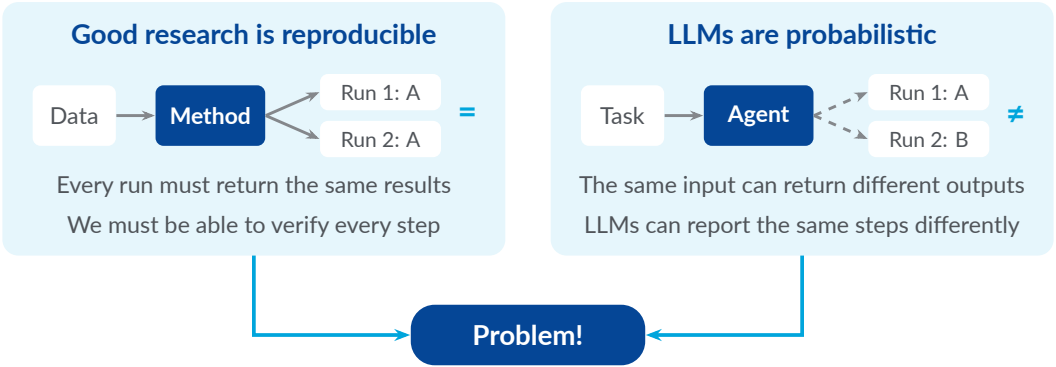
A survey by a software company, with unweighted results, on office work

Where on this scale do I want to be?



Reflect yourself while I explain my current position

One fundamental challenge



To solve this, we need to understand a few basics about AI tooling first

Harness	The software around an LLM that lets it read files, run code, and search the web (e.g., Claude Code, Codex, dsh, and PI)
Agent	An LLM in a harness that works on a task independently and chooses its own steps
Skill	A folder with written instructions and scripts that tell an agent how to do a specific task
AGENTS.md (CLAUDE.md)	A text file with our rules that the agent reads at the start of every session
Other tooling	MCP, evals, hooks, memory, context, sandboxes, mods

What should this AI tooling do?

1

Clear objectives and processes

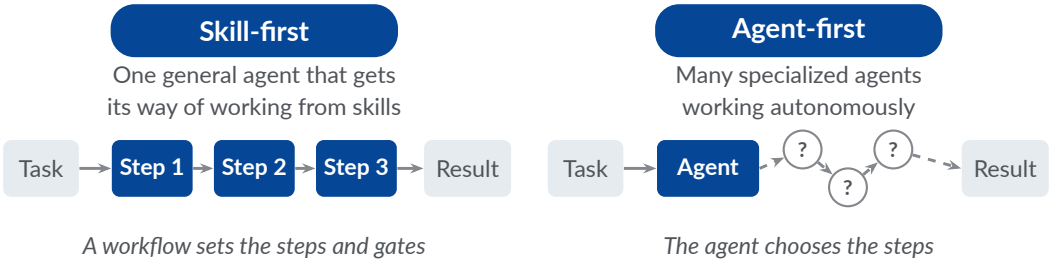
It should help us set well-defined goals and structured processes, which is what reproducible research requires: A clear methodological approach.

2

Clear roles for humans and agents

It should optimize where human judgment is needed/valuable and where boring steps can be delegated to agents (i.e., no need for verification).

My preference: skill-first over agent-first AI tooling

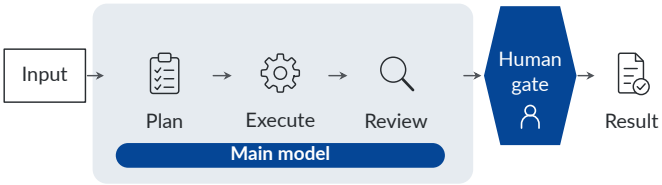


Skill-first tooling can ensure: researcher-in-the-loop.

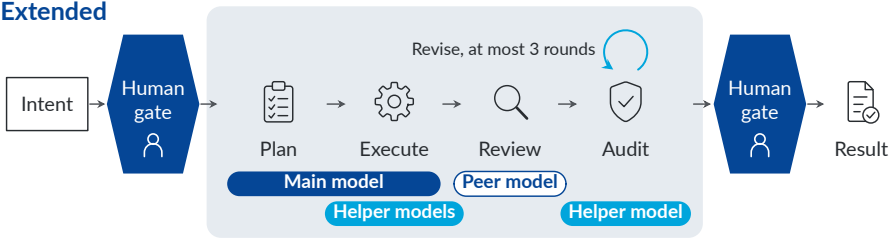
The two are not mutually exclusive: a skill calls further agents for routine steps.

Skill workflows (examples)

Simple



Extended



See the Anthropic documentation and agentskills.io.

Principles for my setup

One general agent, many skills
The skill that matches the task tells the agent how to work

Agents execute, and I verify
A step is handed over in full only when its result needs no verification

Judgment stays with the main model and with me
Helper models receive routine work only

Rules that must always hold are enforced by code
A model can forget a written rule, whereas a script that blocks an action cannot

The simplest mechanism comes first
A line in AGENTS.md, then a skill, then a workflow with several models

Simple example: the paper audit skill

Purpose: Execute an audit (mistakes, errors, inconsistencies, and fact-check) in a paper before it goes to a journal.

- **Input:** I give it the paper, any other materials, and optional instructions
- **Plan:** The main model lists the tables, figures, and references, and the most likely errors
- **Execute:** Four helper models each check one part (references, tables, text, response letter)
- **Review:** The main model reviews the findings, looks for missed errors, and writes one report
- **Audit:** A helper model checks each finding against the paper, and the main model revises
- **Result:** I check the final report, which ranks each mistake with its location

More complex example: the paper reviser skill

Purpose: Execute a preliminary revision of a paper based on comments.

- **Input:** I give it the paper, reviewer/editor/discussant comments, and optional instructions
- **Intent:** The main model writes how it reads my request, and I correct it (1st gate)
- **Plan:** The main model turns each comment into a task and sets the order of the changes
- **Execute:** The main model edits the paper and drafts preliminary responses to the comments, and a helper model verifies new references
- **Review:** A peer model reviews the revisions with fresh eyes, and the main model corrects
- **Audit:** A helper model checks that each comment is addressed correctly
- **Result:** I process each change, the response, and points left open (2nd gate)

“Starter” skills repository

Simple starter skills for writing a research paper and conducting research

Paper finder	Careful search for research papers on a topic
English editor	Corrects grammar and style, and marks every change
Feedback reviser	Processes changes based on comments
Document auditor	Checks every reference, fact, and number in a document

github.com/victorvanpelt/research-skills



But how can AI help us analyze data?

Project folder

- **0_data**
  - raw_data.xxx
  - codebook.md
- **1_code**
  - code.xxx
- **2_process**
  - processed_data.xxx
- **3_output**
  - table_1.xxx
  - figure_1.xxx
- **4_drafts**
  - paper.qmd

Four rules

1. Never modify **0_data**
2. Only save data to **2_process**
3. Only load data from **0_data** or **2_process**
4. Anything in **2_process** and **3_output** can be deleted and regenerated from **1_code**

Agents have two roles:

Role 1: Agents edit the code, or help us edit it

Role 2: AI as a source of generated data only

Source: github.com/victorvanpelt/research_pipeline_example

AGENTS.md: rules for every agent

```
# AGENTS.md (excerpt, shortened)
- Never modify the raw data in 0_data/.
- Never edit 2_process/ or 3_output/ by hand.
- Change only 1_code/, 4_drafts/, and the codebook.
- Before a multi-file change, say which files will change.
- Never run git commit or git push.
- Done only when "make clean" then "make" rebuilds everything.
- Report what changed and anything you could not verify.
```

Git is a quick way to see which changes AI agents made (researcher-in-the-loop)

Source: github.com/victorvanpelt/research_pipeline_example

Recent development: System One models (e.g., Jev)

A new development inspires a classification of models into two types, based on Kahneman's (2011) dual-system theory.

System Two model



Good at complex tasks that need reasoning

System One model



Good at fast and structured decisions

System One models could be useful for research

In a chatbot experiment, we tell the chatbot how to behave in each condition, but we cannot be sure that every reply follows this instruction.

Thus, we let Jev classify how the chatbot behaves in each reply, either to create a quality score after the experiment or even to intervene before the participant sees the reply.

Test on 802 synthetic chatbot replies: how often did Jev answer correctly?

Instruction that Jev checked	Replies that followed it	Replies that did not follow it	All replies
Rule for all participants No markdown or formatting	99.1% 644 of 650	100% 152 of 152	99.3% 796 of 802
Assertive treatment Direct and confident, without hedge words	92.5% 371 of 401	100% 401 of 401	96.3% 772 of 802
Non-assertive treatment Tentative and cautious, with hedge words	98.0% 393 of 401	99.8% 400 of 401	98.9% 793 of 802

Final comments on AI use in research

1

Data protection

We should prefer a local LLM, then a cloud LLM with zero data retention (ZDR) in the EU, then other cloud LLMs. Some data cannot be shared with a cloud LLM at all.

2

Self-disclosure of AI use

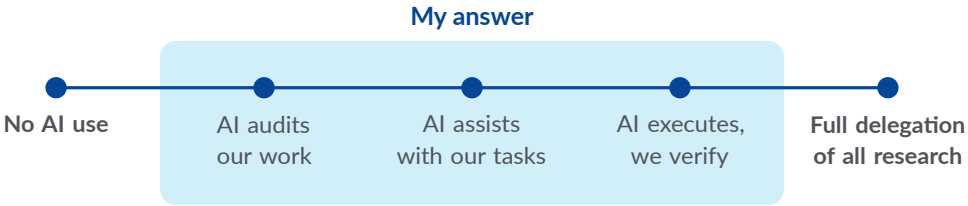
Most journals already require disclosure. It should be standard practice that we reflect on and are transparent about how we use AI, because this prevents misconceptions and promotes responsible use.

3

Statistical inference

AI lowers the hurdles to run many non-hypothesized tests, fish for results, and exploit researcher degree's of freedom. Thus, preregistration and sharing research materials become more important than ever.

My position on the scale



Where on the scale are you / do you want to be?

Thank you for your attention

Slides: **https://victorvanpelt.github.io/ai-research-talk**

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